

UAV-Enabled Relay Communication Under Malicious Jamming: Joint Trajectory and Transmit Power Optimization

Yang Wu¹, Weiwei Yang¹, Xinrong Guan¹, *Member, IEEE*,
and Qingqing Wu¹

Abstract—In this letter, the UAV-enabled relay communication is investigated, where the UAV relay forwards the information received from the ground source node to the ground destination node under malicious jamming. We aim to maximize the end-to-end throughput by optimizing the trajectory of UAV and transmit power of both UAV and source node jointly. However, the formulated problem is intractable due to the non-convex objective function and constraints. Thus, with the aid of slack variables, we propose an efficient algorithm to solve it based on the block coordinate descent (BCD) and successive convex approximation (SCA) techniques. Numerical results are provided to show that the joint design of trajectory and power control improves the throughput significantly.

Index Terms—UAV communication, relay, anti-jamming, trajectory design.

I. INTRODUCTION

Unmanned aerial vehicle (UAV)-enabled communication has attracted increasing interest recently with the advantages of line-of-sight (LoS) channels and highly controllable maneuverability [1]. The UAV can be employed as the base station to provide data dissemination or as the aerial relay to ferry the information from the source node to the destination node [2]. Via jointly optimizing the trajectory and communication resources, such as transmit power, user scheduling and subcarriers, the system throughput can be improved significantly [3], [4].

However, the broadcasting nature of wireless channels makes UAV communications particularly vulnerable to security threats [5], which thus makes that the joint design in [3], [4] can not be directly applied in the presence of malicious eavesdroppers or jammers. To combat the eavesdropping, the trajectories of the information UAV and the friendly jamming UAV were jointly optimized and thus the energy efficiency for secure UAV communication was maximized in [6]. As for jamming, it is conventionally handled in the power domain, frequency domain and/or time domain. Owing to the UAV's controllable trajectory, it is also appealing to combat the jamming in the spatial domain, i.e., retreating away from the jammer [7], [8]. In [9], the full-duplex mode was adopted

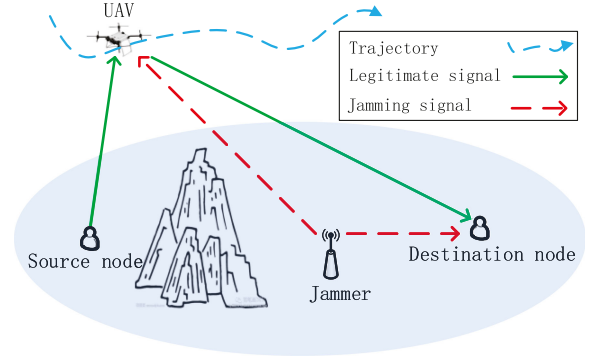


Fig. 1. UAV-enabled relay under malicious jamming.

for the UAV-enabled relay communication with the interference generated by the ground communications, which, however, will cause severe self-interference that requires employing advanced self-interference cancellation technologies. In [10], the signal-to-interference-plus-noise ratio (SINR) of the destination node was optimized via trajectory design to combat the jamming. Nevertheless, the transmissions from the ground source node to the UAV and that from the UAV to the ground destination node were carried out alternatively with equal time slots and the transmit power of the UAV and the source node were both fixed. To our best knowledge, the joint trajectory and transmit power design for UAV-enabled relay communication under malicious jamming is still an open problem.

In this letter, we investigate the UAV-enabled relay communication under malicious jamming, which considers the trajectory of the UAV, the transmit power of the UAV and the source node. We propose an alternating optimization based algorithm to solve the formulated non-convex problem efficiently, wherein the original problem is divided into two sub-problems which optimize the trajectory and transmit power, respectively. Numerical results show that the proposed algorithm can enhance the destination node's throughput significantly especially when the flight time is limited.

II. SYSTEM MODEL AND PROBLEM FORMULATION

We consider a UAV-enabled relay communication system consisting of a ground source node S, a ground destination node D and a UAV relay U, while a jammer M aims to jam the transmission from S to U and U to D as illustrated in Fig. 1. The trajectory and transmit power control will be predesigned in the UAV or a centralized ground controller and thus the source node and the UAV can execute the relay task with the preplanned strategies. We assume that U flies at a fixed minimum altitude H to avoid ground obstacles such as mountains and skyscrapers. Three-dimensions (3D) Cartesian coordinate system is considered. Thus, the locations of S, D, U and M are denoted by $\mathbf{q}_s = [x_s, y_s, 0]^T \in \mathbb{R}^{3 \times 1}$, $\mathbf{q}_d = [x_d, y_d, 0]^T \in \mathbb{R}^{3 \times 1}$, $\mathbf{q}_u = [x_u, y_u, H]^T \in \mathbb{R}^{3 \times 1}$ and $\mathbf{q}_m = [x_m, y_m, 0]^T \in \mathbb{R}^{3 \times 1}$. U is deployed to fly from the start point $\mathbf{q}_u^{\text{start}} = [x_u^{\text{start}}, y_u^{\text{start}}, H]^T \in \mathbb{R}^{3 \times 1}$ to the endpoint $\mathbf{q}_u^{\text{end}} = [x_u^{\text{end}}, y_u^{\text{end}}, H]^T \in \mathbb{R}^{3 \times 1}$ to execute the communication task within a given time period T . For ease of UAV trajectory optimization, the continuous T is divided into N equal time slots d_t so that $T = Nd_t$. Therefore, the trajectory of U can be denoted approximately by the combination of discrete positions, i.e., $\mathbf{q}_u[n] = [x_u[n], y_u[n], H]^T$.

Manuscript received March 29, 2021; revised May 7, 2021; accepted June 10, 2021. Date of publication June 14, 2021; date of current version August 13, 2021. This work was supported by the Natural Science Foundations of China under Grants 61771487 and 61471393. The work of Qingqing Wu was supported in part by the Guangdong NSF under Grant 2021A1515011900, FDCT under Grants 0108/2020/A, 0119/2020/A3, and SKL-IOTSC-2021-2023. The review of this article was coordinated by Dr. X. Cao. (*Corresponding author: Weiwei Yang.*)

Yang Wu, Weiwei Yang, and Xinrong Guan are with the College of Communications Engineering, Army Engineering University of PLA, Nanjing 210007, China (e-mail: wuyang0710@163.com; wwyang1981@163.com; geniusg2017@gmail.com).

Qingqing Wu is with the State Key Laboratory of Internet of Things for Smart City, University of Macau, Macao 999078, China (e-mail: qingqingwu@um.edu.mo).

Digital Object Identifier 10.1109/TVT.2021.3089158

Considering the mobility capability of the UAV, we have the following constraints

$$\mathbf{q}_u[1] = \mathbf{q}_u^{start}, \quad (1)$$

$$\mathbf{q}_u[n] = \mathbf{q}_u^{end}, \quad (2)$$

$$\|\mathbf{q}_u[n] - \mathbf{q}_u[n-1]\| \leq V_{\max} d_t, n = 2, 3, \dots, N, \quad (3)$$

where $V_{\max}$ denote the maximum flying speed of U. Let $P_s[n]$ and $P_u[n]$ denote the transmit power of S and U in time slot n , respectively. Then, we have the following constraints

$$\sum_{n=1}^N d_t p_s[n] \leq E_s, \forall k, \quad (4)$$

$$P_s[n] \leq P_{\text{peak}}, \forall k, n, \quad (5)$$

$$\sum_{n=1}^N d_t p_u[n] \leq E_u, \forall k, \quad (6)$$

$$P_u[n] \leq P_{\text{peak}}, \forall k, n, \quad (7)$$

where E_s and E_u are the total transmit energy available for S and U, respectively.

The field trial have demonstrated that the air-to-ground (A2G) channel is mainly dominated by the large scale path-loss [11]. Hence, the channel power gains from S to U, U to D and M to U can be denoted as $g_{s,u}[n] = \beta_0 \|\mathbf{q}_u[n] - \mathbf{q}_s\|^{-2}$, $g_{u,d}[n] = \beta_0 \|\mathbf{q}_u[n] - \mathbf{q}_d\|^{-2}$ and $g_{m,u}[n] = \beta_0 \|\mathbf{q}_u[n] - \mathbf{q}_m\|^{-2}$, respectively, where β_0 is the channel power gain at the reference distance $d_0 = 1$ m. Moreover, by considering the worst case, the channel power gain from M to D can also be denoted as $g_{m,d} = \beta_0 \|\mathbf{q}_m - \mathbf{q}_d\|^{-2}$. Thus, the achievable throughput from S to U and U to D in time slot n is given by

$$R_{s,u}[n] = d_t \log_2 \left(1 + \frac{P_s[n] g_{s,u}[n]}{P_m g_{m,u}[n] + \sigma^2} \right), \quad (8)$$

and

$$R_{u,d}[n] = d_t \log_2 \left(1 + \frac{P_u[n] g_{u,d}[n]}{P_m g_{m,d} + \sigma^2} \right), \quad (9)$$

respectively, where P_m and σ^2 denote the transmit power of M and the power of additive white Gaussian noise (AWGN), respectively. We assume that U adopts the decode-forward (DF) relaying in frequency division duplexing (FDD) mode, for which U just forwards the data that has been successfully decoded. Thus, the total number of information bits that have been forwarded from U to D should be less than that U have received from S at any time. Then, we have the information-causality constraints as

$$\sum_{n=2}^J R_{u,d}[n] \leq \sum_{n=1}^{J-1} R_{s,u}[n], J = 2, 3, \dots, N. \quad (10)$$

Aiming at maximizing the throughput of the S-D link by jointly optimizing the trajectory of U $\mathbf{q}_u[n]$, the transmit power of S $P_s[n]$ and the transmit power of U $P_u[n]$, the problem is mathematically written as

$$(P1) : \max_{\mathbf{q}_u[n], P_s[n], P_u[n]} \sum_{n=1}^N R_{u,d}[n] \quad (11)$$

s.t. (1) – (7), (10).

Note that problem (P1) is difficult to solve since the constraint (10) and the objective function (11) are non-convex, while the optimization variables are coupled with each other. Then, by applying the BCD

method, we decompose problem (P1) into two sub-problems which optimize the trajectory $\mathbf{q}_u[n]$, the transmit power $P_s[n]$ and $P_u[n]$, respectively. A suboptimal solution can be obtained by alternatively solving the two sub-problems in an iterative manner until the algorithm converges.

III. PROPOSED ALGORITHM

A. Transmit Power Optimization

Given UAV trajectory, problem (P1) can be rewritten as

$$(P3) : \max_{P_s[n], P_u[n]} \sum_{n=1}^N R_{u,d}[n] \quad (12)$$

$$\text{s.t. (4) – (7), (10).}$$

Note that the term $R_{u,d}[n]$ in the left-hand-side (LHS) of (10) is non-convex with respect to $P_u[n]$, we use the following lemma to transform $R_{u,d}[n]$ into convex.

Lemma 1: For any given feasible point $P_u^f[n]$, $R_{u,d}[n]$ is upper bounded by

$$R_{u,d}^{up}[n] = \log_2(1 + P_u^f[n] \gamma_{u,d}[n]) + \log_2 e \frac{\gamma_{u,d}[n]}{1 + P_u^f[n] \gamma_{u,d}[n]} (P_u[n] - P_u^f[n]), \forall n, \quad (13)$$

where $\gamma_{u,d}[n] = \frac{g_{u,d}[n]}{P_m g_{m,u}[n] + \sigma^2}$ is the SINR of D.

Proof: Since $f(x) = \log_2(1 + ax)$ is concave with respect to x , at a given feasible point x^f , a global over-estimator can be provided with the aid of its first-order Taylor expansion

$$\log_2(1 + ax) \leq \log_2(1 + ax^f) + \log_2 e \frac{a}{1 + ax^f} (x - x^f). \quad (14)$$

Hence, by letting $x = P_u$ and $a = \gamma_{u,d}$, Lemma 1 is proved. ■

According to Lemma 1, (10) can be reformulated as

$$\sum_{n=2}^J R_{u,d}^{up}[n] \leq \sum_{n=1}^{J-1} R_{s,u}[n], J = 2, 3, \dots, N. \quad (15)$$

Then, problem (P3) can be transformed into

$$(P4) : \max_{P_s[n], P_u[n]} \sum_{n=1}^N R_{u,d}[n] \quad (16)$$

$$\text{s.t. (4) – (7), (15).}$$

Problem (P4) is a standard convex optimization problem with respect to $P_s[n]$ and $P_u[n]$. Hence, it can be efficiently solved with CVX [12]. Moreover, its solution is guaranteed to be the feasible solution for problem (P3).

B. Trajectory Optimization

Given transmit power $P_s[n]$ and $P_u[n]$, problem (P1) can be rewritten as

$$(P5) : \max_{\mathbf{q}_u[n]} \sum_{n=1}^N R_{u,d}[n] \quad (17)$$

$$\text{s.t. (1) – (3), (10).}$$

Noting that the term $R_{u,d}[n]$ in the objective function, $R_{s,u}[n]$ and $R_{u,d}[n]$ in the constraint (10) are neither concave nor convex with

respect to $\mathbf{q}_u[n]$, which makes (P5) intractable. Thus, a slack variable $t[n]$ is introduced, which satisfies

$$x[n]^2 + y[n]^2 + H^2 \leq t[n], \forall n. \quad (18)$$

As a result, problem (P5) can be transformed into an equivalent problem

$$(P6): \max_{\mathbf{q}_u[n], t[n]} \sum_{n=1}^N \tilde{R}_{u,d}[n] \quad (19)$$

$$\text{s.t.} \quad \sum_{n=2}^J \tilde{R}_{u,d}[n] \leq \sum_{n=1}^{J-1} R_{s,u}[n], J=2, 3, \dots, N, \quad (20)$$

$$(1) - (3), (18),$$

where

$$\tilde{R}_{u,d}[n] = d_t \log_2 \left(1 + \frac{P_u[n]\beta_0}{P_m g_{m,d} + \sigma^2} \frac{1}{t[n]} \right). \quad (21)$$

The equivalence can be proved by contradiction. Specifically, when we obtain the optimal solution, if (18) holds with inequality, the value of $t[n]$ can always be decreased to improve the objective function's value.

With the help of slack variable $t[n]$, we transform the LHS of (20) into convex, however, the objective function is still non-concave. Then, the following lemma is introduced to reformulate the objective function into concave.

Lemma 2: For any given feasible point $t^f[n]$, $\tilde{R}_{u,d}[n]$ is lower bounded by

$$\begin{aligned} \tilde{R}_{u,d}^l[n] &= d_t \log_2 \left(1 + \frac{P_u[n]\beta_0}{P_m g_{m,d} + \sigma^2} \frac{1}{t^f[n]} \right) \\ &+ d_t \log_2 e \left(\frac{-P_u[n]\beta_0}{P_u[n]\beta_0 t^f[n] + P_m g_{m,d} + \sigma^2 t^f[n]^2} \right) (t[n] - t^f[n]). \end{aligned} \quad (22)$$

Proof: Since $f(x) = \log_2(1 + \frac{a}{x})$ is a convex function with respect to x , at a given feasible point x^f , a global under-estimator can be provided with the aid of its first-order Taylor expansion

$$\log_2 \left(1 + \frac{a}{x} \right) \geq \log_2 \left(1 + \frac{a}{x^f} \right) + \log_2 e \frac{-a}{a x^f + x^{f2}} (x - x^f). \quad (23)$$

Hence, by letting $x = t[n]$ and $a = \frac{P_u[n]\beta_0}{P_m g_{m,d} + \sigma^2}$, Lemma 2 is proved. ■

According to Lemma 2, we transform the objective function into concave and reformulate problem (P6) as

$$(P7): \max_{\mathbf{q}_u[n], t[n]} \sum_{n=1}^N \tilde{R}_{u,d}^l[n] \quad (24)$$

$$\text{s.t.} \quad (1) - (3), (18), (20).$$

Then, to transform the right-hand-side (RHS) of (20) into concave, two slack variables $L_{s,u}[n]$ and $I_u[n]$ are introduced such that

$$P_s[n]g_{s,u}[n] \geq L_{s,u}[n]^{-1}, \forall n, \quad (25)$$

$$P_m g_{m,u}[n] + \sigma^2 \leq I_u[n], \forall n. \quad (26)$$

Thus, we further transform problem (P7) into an equivalent problem as

$$(P8): \max_{\mathbf{q}_u[n], t[n], L_{s,u}[n], I_u[n]} \sum_{n=1}^N \tilde{R}_{u,d}^l[n] \quad (27)$$

$$\begin{aligned} \text{s.t.} \quad & \sum_{n=2}^J \tilde{R}_{u,d}[n] \\ & \leq \sum_{n=1}^{J-1} d_t \log_2 \left(1 + \frac{1}{L_{s,u}[n] I_u[n]} \right), J=2, 3, \dots, N. \end{aligned} \quad (28)$$

$$(1) - (3), (18), (25), (26).$$

The equivalence can be proved by contradiction. Specifically, if (25) and (26) hold with inequalities in the optimal solution, the value of $L_{s,u}[n]$ and $I_u[n]$ can always be decreased to improve the objective value.

Then, we use the following lemma to obtain a lower bound of RHS of (28) which is concave.

Lemma 3: For any given feasible point $(L_{s,u}^f[n], I_u^f[n])$, $\log_2(1 + \frac{1}{L_{s,u}[n] I_u[n]})$ is lower bounded by

$$\begin{aligned} R_{s,u}^l[n] &= \log_2(1 + 1/L_{s,u}^f[n] I_u^f[n]) \\ &+ A_{s,u}(L_{s,u}[n] - L_{s,u}^f[n]) + B_{s,u}(I_u[n] - I_u^f[n]), \end{aligned} \quad (29)$$

where $A_{s,u} = -\log_2 e / (L_{s,u}^f[n] + (L_{s,u}^f[n])^2 I_u^f[n])$ and $B_{s,u} = -\log_2 e / (I_u^f[n] + (I_u^f[n])^2 L_{s,u}^f[n])$.

Proof: Since $f(x, y) = \log_2(1 + 1/xy)$ is a convex function with respect to x and y , at a given feasible point (x^f, y^f) , a global under-estimator can be provided with its first-order Taylor expansion, i.e.,

$$\begin{aligned} \log_2(1 + 1/xy) &\geq \log_2(1 + 1/x^f y^f) \\ &- (x - x^f) \log_2 e / (x^f + (x^f)^2 y^f) \\ &- (y - y^f) \log_2 e / (y^f + (y^f)^2 x^f). \end{aligned} \quad (30)$$

Hence, by letting $x = L_{s,u}[n]$ and $y = I_u[n]$, Lemma 3 is proved. ■

However, (26) is still non-convex. Then, slack variable $d_m[n]$ is introduced to reformulate (26) as the following equivalent constraints

$$\sum_{m=1}^M P_m \beta_0 d_m[n]^{-1} + \sigma^2 \leq I_u[n], \forall n, \quad (31)$$

$$d_m[n] \leq \|\mathbf{q}_u[n] - \mathbf{q}_m\|^2, \forall n, \quad (32)$$

$$d_m[n] \geq 0, \forall n. \quad (33)$$

The equivalence can also be proved by contradiction that if the optimal solution is obtained, (32) should hold with equality, otherwise, the value of $d_m[n]$ can always be increased to enhance the objective function's value.

Noting that (32) is non-convex, with the first-order Taylor expansion of $\|\mathbf{q}_u[n] - \mathbf{q}_m\|^2$, the lower bound of $\|\mathbf{q}_u[n] - \mathbf{q}_m\|^2$ can be obtained as

$$\begin{aligned} q_{u,m}^l[n] &= 2x^f[n]x[n] - (x^f[n])^2 + x_m^2 - 2x_m x[n] \\ &+ 2y^f[n]y[n] - (y^f[n])^2 + y_m^2 - 2y_m y[n], \end{aligned} \quad (34)$$

with which (32) can be reformulated into a convex constraint

$$d_m[n] \leq q_{u,m}^l[n]. \quad (35)$$

Finally, problem (P8) is reformulated as

$$(P9): \max_{\mathbf{q}_u[n], t[n], L_{s,u}[n], I_u[n], d_m[n]} \sum_{n=1}^N \tilde{R}_{u,d}^l[n] \quad (36)$$

$$\text{s.t.} \quad \sum_{n=2}^J \tilde{R}_{u,d}[n] \leq \sum_{n=1}^{J-1} R_{s,u}^l[n], J=2, 3, \dots, N, \quad (37)$$

$$(1) - (3), (18), (25), (31), (33), (35).$$

Noting that problem (P9) is a standard convex optimization problem and thus it can be efficiently solved with CVX [12].

Algorithm 1: The proposed algorithm for solving problem (P1).

- 1: **Initialization:** Denote the feasible solution in i_{th} iteration as $\mathbf{q}_u^i[n]$, $P_s^i[n]$ and $P_u^i[n]$, set $i = 0$ as the initial feasible point.
 - 2: **Repeat**
 - 3: With given $\mathbf{q}_u^i[n]$, update $P_s^i[n]$ and $P_u^i[n]$ to $P_s^{i+1}[n]$ and $P_u^{i+1}[n]$ by solving problem (P4).
 - 4: With given $P_s^{i+1}[n]$ and $P_u^{i+1}[n]$ update $\mathbf{q}_u^i[n]$ to $\mathbf{q}_u^{i+1}[n]$ by solving problem (P9).
 - 5: Update $i = i + 1$.
 - 6: **Until** The increasement of the objective value is smaller than the threshold μ .
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C. Overall Algorithm

To summarize, by introducing the BCD method, we propose an efficient algorithm that decomposes the original problem into two sub-problems which can be solved in an iterative way with the SCA technique. The iterative loop ends when the increasement of the objective function's value is smaller than a threshold μ . Moreover, the algorithm is guaranteed to converge. This is because the solution of each sub-problem is non-decreasing in each iteration, and the value of original problem is finite for the throughput of the communication system cannot be unbounded. Note that each iteration of Algorithm 1 requires solving the convex optimization problem (P4) and (P9), which has a polynomial complexity $O((x)^{3.5})$ in the worst case, where x is the number of variables. Thus, the complexity of Algorithm 1 is $O(k(8N)^{3.5})$, where k is the number of iterations. The specific algorithm are summarized in Algorithm 1.

IV. NUMERICAL RESULTS

Numerical results are provided to verify the performance of Algorithm 1 that optimize the trajectory and transmit power jointly (denoted as "Joint optimization"). Particularly, two benchmark algorithms are adopted: 1) The UAV-enabled relay flies with line segment trajectory and fixed transmit power (denoted as "Line trajectory"), and 2) The trajectory and transmit power are jointly optimized without considering jamming signals (denoted as "Joint w/o J"). The related parameters are set as follows. The duration of each time slot $d_t = 0.5$ s and the noise power is $\sigma^2 = -119$ dBm. The peak transmit power for source node S and UAV are $P_{peak} = 0.4$ W. The transmit power of jammer M is $P_m = 0.1$ W. The altitude of the level flight of UAV is $H = 100$ m. The maximum speed of UAV is $V_{max} = 60$ m/s. The channel power gain at the reference distance $d_0 = 1$ m is $\beta_0 = -60$ dB. The convergence threshold is $\mu = 10^{-3}$. Without loss of generality, S, D and M are placed in (0, 1000, 0)m, (1000, 1000, 0)m and (500, 500, 0)m respectively. The startpoint and endpoint of U are (0, 0, H)m and (1000, 0, H)m, respectively.

Fig. 2 shows the trajectories of the UAV for different algorithms when $E_s = E_u = 50$ joule. Particularly, in the "Line optimization" algorithm, if the flight time is sufficiently long, the UAV will fly through S and D, otherwise, its trajectory will be a triangle that approaches S and D as close as possible. It is observed that in all algorithms when $T = 60$ s, the UAV first flies from the startpoint to S and then turns to D and finally reaches the endpoint. Specifically, the UAV tends to fly away from M when approaching and leaving S in the "Joint optimization" algorithm while the UAV tends to fly close to M when approaching S and endpoint in the "Joint w/o J" algorithm. This is because the further distance from M can provide better channel quality by retreating away from jamming signals in the "Joint optimization" algorithm while the

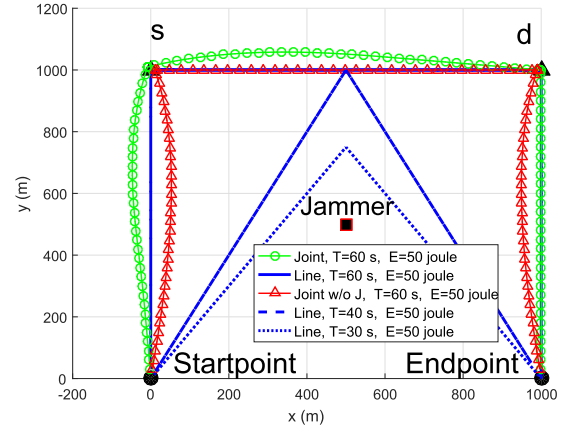


Fig. 2. Trajectories of UAV in three algorithms.

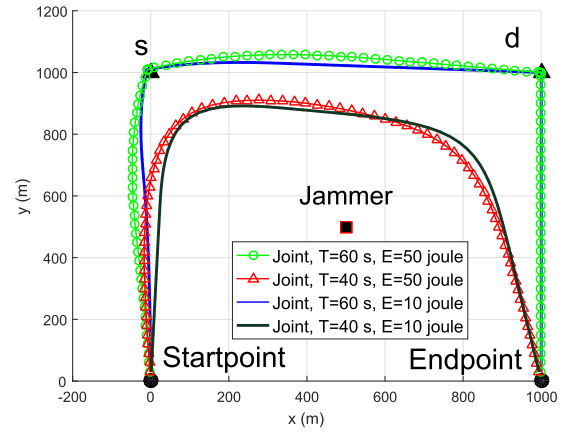


Fig. 3. Trajectories of UAV in "Joint optimization" algorithm with different T and E .

closer distance to both S and D (also closer to M in this case) can also provide better channel quality if there are no jamming signals in the "Joint w/o J" algorithm.

Fig. 3 shows the trajectories of the UAV in "Joint optimization" algorithm with different T and E . It is observed that the trajectory of the UAV is closer to S than D when $T = 60$ s as compared to the trajectory with $T = 40$ s. This is because given sufficient flight time, a closer distance to S or D can both improve the channel quality from S to U or U to D and reduce the channel quality from M to U. Meanwhile, it is observed that with larger available energy E , the trajectory of UAV with the same T tends to fly further away from M. This is because although the transmissions from S to U carry out in the location with best channel quality (hovering near above S) is optimal, the transmissions conducted there are restricted by the peak transmit power constraint. Thus, with more available energy E , some transmissions have to be conducted when the UAV is far from S and close to M. Then, the UAV has to keep a further distance to M so as to combat the jamming.

Fig. 4 shows the throughput of U and D over the time slot for different algorithms with $T = 60$ s and $E = 50$ joule. It is observed that both the throughput of U and D increases slowly in the first and then increases rapidly in "Joint optimization" algorithm. This is because both S and U tend to reduce the transmit power when the UAV is far from S or D and enhance the transmit power when the UAV approaches S or D. Meanwhile, the "Line optimization" algorithm violates the information

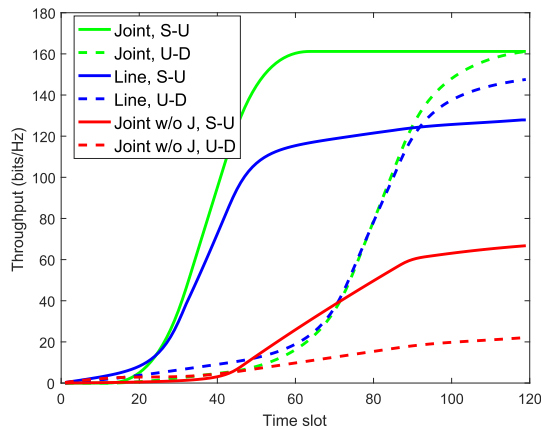
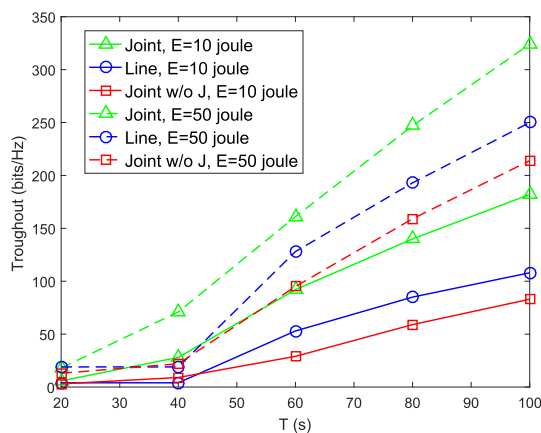


Fig. 4. Throughput of U and D versus time slot.

Fig. 5. Throughput of D with different T .

information-causality constraint significantly in the latter time period as the throughput of U exceeds the throughput of D for its fixed trajectory and transmit power control. The “Joint w/o J” algorithm achieves much smaller throughput of D compared to both “Joint optimization” and “Line optimization” algorithms for not taking jamming signals into account.

Fig. 5 shows the throughput of D with different T in the three algorithms when $E = 10$ joule and $E = 50$ joule, respectively. It is observed that the throughput of D in all algorithms increase over time, and the “Joint optimization” algorithm always performs better than the “Line optimization” and “Joint w/o J” algorithms. Particularly, “Joint optimization” algorithm is more necessary when the flight time T is small, e.g., the throughput of “Joint optimization” algorithm when

$T = 40$ s is more than three times of that of the “Line optimization” and “Joint w/o J” algorithms, while the throughput of “Joint optimization” algorithm when $T = 100$ s is less than two times of that of the “Line optimization” and “Joint w/o J” algorithms. Thus, it is verified that the joint trajectory design and transmit power control can improve the throughput especially when the flight time is limited.

V. CONCLUSION

In this letter, we studied the UAV-enabled relay communication under malicious jamming. A joint trajectory and transmit power optimization algorithm was proposed to solve the formulated non-convex problem. Numerical results verified the effectiveness of the proposed algorithm and show that the joint design of trajectory and transmit power can improve the throughput of destination node significantly especially when the flight time is limited.

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